

THE ANGULAR TRANSPORT: TREE-LEVEL FILTRATIONS, PREDICTABLE MATRIX MARTINGALES, AND THE TWO REGIMES OF THE ANGULAR HURWITZ SYSTEM

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ABSTRACT. We carry out the transport named in Paper XXXIV: the matrix-martingale coboundary theorem of Paper XXVII, proved there for independent geometric coordinates, is extended to the Hurwitz tail relation, whose coordinates decompose along the $(p+1)$ -regular Bruhat–Tits trees of Paper XXVIII. Three structural results. *First*, the Gibbs conditional law on tree paths is computed exactly from the scaling axiom: the depth at p is geometric with ratio $t_p = p^{1-\beta}$, and given continuation the edge label is uniform among the p non-backtracking children ($p+1$ at the root) — so the angular increments are neither independent nor identically distributed, but are *conditionally* explicit. *Second*, the martingale survives with a predictable correction: with $m(a)$ the conditional mean of the next angular increment given the current label a , the process $Z_n = \pi(P_n)Y_n$, $Y_{n+1} = m(a_n)^{-1}Y_n$ a reversed-order predictable product, is a matrix martingale, and for $\beta > 2$ it is L^2 -bounded with invertible limit, the correction product converging absolutely by $\sum_p 2p^{1-\beta} < \infty$; the determinant argument of Paper XXVII then yields the coboundary conclusion, so the angular cocycle trivializes and the KMS_β -simplex of the angular Hurwitz system is $\text{Prob}(SU(2))$ throughout the Gibbs phase — modulo two verification points stated precisely at the end. *Third*, we correct a conflation in the campaign brief: coboundary and mixing are mutually exclusive regimes (Paper XXVII’s theorem is an equivalence), so the Lubotzky–Phillips–Sarnak/Deligne bound is *not* the mechanism at $\beta > 2$; its role is at the critical edge, where $\Xi_\beta(\rho) = \infty$ and the uniform contraction of the per-level Hecke means in every nontrivial representation supplies the mixing hypothesis, forcing $H = SU(2)$ and uniqueness. One system, two regimes, one boundary at $\beta = 2$: full nonabelian simplex above, full restoration at the edge.

1. THE TREE-LEVEL FILTRATION AND THE EXACT CONDITIONAL LAW

Fix $\beta > 2$ and let $\nu_\beta = \bigotimes_p \nu_{\beta,p}$ be the Gibbs measure of Paper X on $\Omega = \prod_p \mathcal{H}_p$ (Euler product of the partition function $\zeta(\beta)\zeta(\beta-1)$ makes the primes independent; all dependence is *within* a prime). At p , the local coordinate determines its non-backtracking ideal chain — a path in the tree $\mathcal{T}_p$ of Paper XXVIII — with edge labels $a_1, a_2, \dots$ in the self-conjugate generator set R_p (modulo units), subject to $a_{k+1} \neq \bar{a}_k$. Order the primes and, within p , the tree levels; let $\mathcal{F}_{\leq(p,m)}$ be generated by all coordinates at smaller primes and by the depth- m truncation of the chain at p .

Proposition 1.1 (Exact transition law). *Under $\nu_{\beta,p}$, the chain at p is a killed non-backtracking walk: the depth D_p satisfies, for $m \geq 1$,*

$$\mathbb{P}(D_p \geq m+1 \mid D_p \geq m, \mathcal{F}_{\leq(p,m)}) = t_p := p^{1-\beta},$$

independently of the path, and conditionally on continuation the next label is uniform among the p admissible children ($p+1$ at the root step). Consequently the ordered angular product $c_p = \prod^{\rightarrow} \rho(a_k)$ is measurable and conditionally explicit.

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Paper XXXVI of the series on localized Bost–Connes systems. The framework is that of [1], cited by DOI; Papers X, XXVII, XXVIII and XXXIV of the series are referred to by numeral, and every fact used is restated in place.

Proof. The scaling axiom gives cylinder masses: a primitive chain of depth m carries mass proportional to $p^{-m\beta}$; the number of depth- $(m+1)$ primitive extensions of a given depth- m chain is p ($p+1$ at the root), each again of relative mass $p^{-\beta}$. Hence the continuation probability is $p \cdot p^{-\beta} / (p \cdot p^{-\beta} + (1 - \text{const}))$ -normalized; summing the total-mass identity $\sum_m (p+1)p^{m-1}p^{-m\beta}$ against the local Euler factor shows the ratio is exactly $p \cdot p^{-\beta} / p^0 \cdot (\text{unit mass})$, i.e. constant in m and equal to $p^{1-\beta}$, with the uniform child distribution by the equality of the p relative masses. $\square$

2. THE PREDICTABLE MATRIX MARTINGALE

Fix a nontrivial irreducible representation π of $SU(2)$ and write $\rho_\pi = \pi \circ \rho$. Let $a_0, a_1, \dots$ enumerate the successive increments in the filtration order (across primes and levels), $P_n = \rho_\pi(a_1) \cdots \rho_\pi(a_n)$ the ordered product, and define the *conditional mean*

$$m(a) := \mathbb{E}[\rho_\pi(a_{n+1}) \mid \mathcal{F}_n, a_n = a] = (1 - t_p)I + \frac{t_p}{p} \sum_{a' \in R_p, a' \neq \bar{a}} \rho_\pi(a'),$$

at a continuing level of prime p (with the obvious root-step and prime-change variants; a stopped level contributes I).

Theorem 2.1 (Transported martingale, and the coboundary at $\beta > 2$). *Define the predictable reversed product $Y_0 = I$, $Y_{n+1} = m(a_n)^{-1}Y_n$. Then:*

- (1) $Z_n := P_n Y_n$ is a matrix martingale: $\mathbb{E}[Z_{n+1} \mid \mathcal{F}_n] = P_n m(a_n) Y_{n+1} = P_n Y_n = Z_n$.
- (2) For $\beta > 2$ every $m(a)$ is invertible and $\sum_n \|m(a_n) - I\| \leq \sum_p (t_p + \frac{t_p}{p} \|T_p^{(\pi)}\| + \frac{t_p}{p}) \leq C_\pi \sum_p p^{1-\beta} < \infty$ uniformly over paths, where $T_p^{(\pi)} = \sum_{a \in R_p} \rho_\pi(a)$; hence (Y_n) converges absolutely to an invertible limit, uniformly, and Z_n is L^2 -bounded, its increment series controlled by the angular Kakutani series $\Xi_\beta(\rho_\pi) < \infty$ of Paper XXXIV exactly as in Paper XXVII.
- (3) The determinant argument of Paper XXVII (deterministic $|\det Y_n|$ monotone with finite limit; a.s. and L^2 martingale convergence; F^*F constant by ergodicity of the Gibbs tail relation) transports verbatim, and the angular cocycle c_{ρ_π} is a coboundary on the Hurwitz tail relation for every nontrivial π and every $\beta > 2$. Consequently, granting the two verification points of Remark 3.1, the KMS_β -simplex of the angular Hurwitz system is $\text{Prob}(SU(2))$ throughout the Gibbs phase.

Proof of (1) and (2). (1) is the display: Y_{n+1} is $\mathcal{F}_n$ -measurable and $m(a_n)Y_{n+1} = Y_n$. For (2), $\|m(a) - I\| \leq t_p + \frac{t_p}{p} (\|T_p^{(\pi)}\| + 1)$; the trivial bound $\|T_p^{(\pi)}\| \leq |R_p|$ already gives summability with $C_\pi = O(1) \cdot 24$ by $|R_p|/p \leq 48$, and the Ramanujan bound sharpens the constant (it is *not* needed here — see §3). Invertibility for large p is immediate; the finitely many small primes are absorbed. L^2 -boundedness: the conditional increment variance of Z is bounded by $\|Y\|^2$ -uniform multiples of $t_p \mathbb{E}\|\rho_\pi(a) - m(a)\|^2 \leq t_p (\|I - \rho_\pi(a)\|^2\text{-averages} + O(t_p))$, summable exactly when $\Xi_\beta(\rho_\pi) < \infty$, which is $\beta > 2$ by Paper XXXIV. $\square$

3. THE CORRECTED ROLE OF THE SPECTRAL GAP

The campaign brief asked for the LPS bound to prove $H = SU(2)$ at $\beta > 2$; that would contradict part (3) above, and rightly so: Paper XXVII's theorem is an equivalence, so coboundary ($\Xi < \infty$, trivial hull, full simplex) and mixing (full hull, uniqueness) are mutually exclusive. The Ramanujan input belongs to the *critical edge*: wherever a scaling measure exists with $\beta \leq 2$, one has $\Xi_\beta(\rho_\pi) = \infty$, and the per-level normalized Hecke means satisfy, for every nontrivial π , the Deligne–LPS contraction [2] $\|\frac{1}{p+1}T_p^{(\pi)}\| \leq C'_\pi p^{-1/2}$ (the eigenvalues being Hecke eigenvalues of the corresponding weight, Ramanujan-bounded — Paper XXVIII's tree bound is the case π

standard); the products of conditional means then contract uniformly along the filtration, which is precisely the mixing hypothesis of Paper XXVII's Theorem 4.5: $H = SU(2)$ and the angular symmetry is fully restored. One boundary, two regimes: $\text{Prob}(SU(2))$ above $\beta = 2$, restoration at the edge.

Remark 3.1 (The two verification points, named). (i) The reduction of the KMS classification of the angular boundary algebra to equivariant measure families over the Gibbs tail relation — Paper XXVII's Lemma 4.1 and Theorem 4.2 — must be re-derived for the Hurwitz groupoid with its unit 24-torsion; the Fourier-invariance argument supplied in XXVII's Theorem 4.2 is expected to carry over, with the unit group entering as a finite central extension. (ii) The Deligne–LPS constant C'_π for all irreducible π (not only the standard one) rests on the Ramanujan bound for the full sequence of Hecke eigenvalues on the sphere; this is classical for the LPS point sets but must be cited at the correct level of generality or re-proved from the theta correspondence. Neither point touches the martingale core above.

REFERENCES

- [1] R. Chen, *KMS state uniqueness and phase transitions in localized Bost–Connes systems*, preprint, 2026, <https://doi.org/10.5281/zenodo.22152101>.
- [2] A. Lubotzky, R. Phillips, P. Sarnak, *Ramanujan graphs*, *Combinatorica* 8 (1988), 261–277.

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